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The semantics of plural morphology in Akan

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With initial native speaker judgments suggesting an exclusive plural morpheme in downward entailing contexts in Akan, as reported by Ahenkorah (2022), this research seeks to test these claims through experimental means and to provide a more comprehensive picture of Akan's plural semantics. The current literature on exclusive and inclusive plurals focuses mainly on English, with some investigation of Western Armenian, Turkish, and Korean (Bale & Khanjian 2008; Renans et al. 2017; Kang 1994: a.o.). These languages have drawn intrigue due to their unmarked form behaving number neutrally, while English's unmarked form is specified for atomicity. Akan, which marks both singulars and plurals morphologically, makes for an interesting addition to this discussion. Ahenkorah (2022) claims that the plural circumfix, used to mark kinship terms, is inclusive, similarly to English, but that the prefix, which is used for most nominals, is exclusive. The present experiments test these two plural morphemes in multiple downward entailing contexts to obtain the full picture of their meanings. The results of the two experiments show that both the plural prefix and plural circumfix have an inclusive reading in downward entailing contexts. Furthermore, this provides evidence that the plural circumfix is not a circumfix at all but is two separate morphemes: a plural prefix and kinship suffix.



1 Introduction

Plural marking in English gives rise to a multiplicity inference, i.e., it has a ‘more than one’ interpretation in positive sentences (Krifka 1989; Sauerland 2003; 2008). For example, (1a) is false in a scenario where John planted only one tree or no trees at all. But this inference disappears in negative sentences, as in (1b).

- (1) a. John planted trees.
 $\rightsquigarrow$ *John planted **more than one** tree.*
 $\neq$ *John planted **a (single)** tree.*
- b. John didn’t plant trees.
 $\neq$ *John did not plant **more than one** tree.*
 $\rightsquigarrow$ *John did not plant **a (single)** tree.*

This difference between the interpretation of the plurals in (1a) and (1b) can also be framed in terms of what the denotation of a plural is in English: It includes both atomic and plural individuals. However, the interpretation of the plural in (1a) is understood as only making reference to pluralities due to a multiplicity inference where the plural is understood as meaning ‘more than one’. In other words, the plural *trees* in (1a) has an **exclusive** interpretation in that it excludes the reference to atomic individual trees. It is only true if John planted more than one tree. But in the negative sentence, the denotation of the plural must include atomic individuals. The plural in (1b) is considered to be **inclusive** due to the fact that it includes both atomic and plural reference. The sentence is true in a situation where John did not plant a single tree and he did not plant a plurality of trees. The inclusive reading of plurals also emerges in downward entailing contexts other than negation, such as under the restrictor of a universal quantifier, shown in (2), as well as in questions (3).¹

- (2) Everyone who planted trees got a gift card.
 $\rightsquigarrow$ *Everyone who planted **one or more** trees got a gift card.*
- (3) Did John plant trees?
 $\rightsquigarrow$ *Did John plant **one or more** trees?*

However, the exclusive reading of *trees* in (1b) is still accessible in a downward entailing sentence if a context, such as the continuation in (4), is incompatible with the inclusive interpretation.

¹ While the debate in the literature as to whether questions are truly downward entailing contexts must be acknowledged, they license NPIs and are subject to the principle of scale reversal as true downward entailing environments are (Guerzoni & Sharvit 2007). Therefore, for the purpose of this research, questions are thus grouped in with the downward entailing contexts of negation and under the scope of universal quantifiers.

In English, a sentence such as (4) is usually pronounced with emphasis on the plural marked noun. While available, this reading is generally marked and considered to be dispreferred.²

(4) John didn't plant **trees**... he only planted one.

In English, plural nouns have been found to have both inclusive and exclusive readings available in downward entailing contexts and only the exclusive reading available in an upward entailing context, as illustrated above. By contrast, in the literature, some languages with different plural marking systems (Western Armenian (Bale & Khanjian 2008; 2014), (controversially) Turkish (Görgülü 2012; Renans et al. 2017), and Korean (Kang 1994; Kim 2005; Kim & Melchin 2018)) are claimed to have exclusively exclusive plurals, meaning that the plural is interpreted as exclusive no matter the entailment pattern of the sentence. Western Armenian, Turkish, and Korean differ from English in that their unmarked form is number-neutral, while English's unmarked form has a strict singular interpretation. Therefore, the plural marked form in Western Armenian, Turkish, and Korean is contrasting not with a singular, but with a number-neutral noun. Furthermore, a fourth language with a claim of an exclusively exclusive plural, Akan differs from both English and these three languages in that it marks both the singular and plural morphologically (Ahenkorah 2022). This yields an opportunity to investigate an additional number marking system in regard to the effect that monotonicity has on the interpretation of plurals in that system.

To test the meaning of plural nouns in Akan, adult native speakers were asked to judge the acceptability of plural nouns in upward and various downward entailing contexts. As Ahenkorah (2022) claims that different morphological markers of plurality can differ in their interpretation with respect to clusivity, both the plural prefix and circumfix were tested in separate experiments. The results of these experiments show that plural nouns in Akan are interpreted inclusively for both plural markers, a finding contradictory to the current literature.

In the next section, Section 2 I discuss the current claims for an exclusively exclusive plural in Western Armenian, Turkish, and Korean, as well as introduce the subject language of this work: Akan. Following this, Section 3 describes the research questions and hypotheses posited for the two experiments. Section 4 presents the methods, results, and discussion of Experiments 1 and 2, respectively. A general discussion summarizes the findings in Section 5, and then the paper is brought to a close with Section 6, the Conclusion.

² A variety of approaches has been proposed to account for the above phenomenon. Each of these approaches is able to account for plural nouns having both inclusive and exclusive interpretations available, with each taking a different perspective. The three approaches are as follows: The ambiguity approach (Farkas & de Swart 2010; Grimm 2013; Martí 2020), the implicature approach (Sauerland et al. 2005; Spector 2007; Zweig 2009), and the homogeneity approach (Križ 2017). These approaches have been tested experimentally in English, Turkish, and Greek (Renans et al. 2017; 2018; 2020; Tieu et al. 2020). This project, however, focuses on first establishing if Akan has inclusive plurals. I leave determining the particular approach to the inclusive interpretation to a future experiment.

2 Current claims for an exclusive plural

Korean, Turkish, and Western Armenian are claimed to have an exclusive plural in downward entailing contexts on account of their number marking pattern. The proposed denotation for the marked / unmarked pairings in these languages is shown in (5), exemplified with Western Armenian *dəgha-ner*, meaning ‘boys’. (5a) is considered number neutral, while the marked noun (5b) can only refer to pluralities.

- (5) In a context where *boys* are *a*, *b*, and *c*:
- a. $\llbracket \text{dəgha} \rrbracket = \{a, b, c, ab, ac, bc, abc\}$
 - b. $\llbracket \text{dəgha-ner} \rrbracket = \{ab, ac, bc, abc\}$

The claims for Korean’s exclusive plural come from Kang (1994); Kim (2005); Kim & Melchin (2018: a.o.). They claim that Korean’s unmarked nouns, such as *sakwa* can mean ‘an apple’ or ‘apples’, while *sakwa-tul* can only be interpreted as ‘apples’, (Kim & Melchin 2018: p.9). However, Bale & Khanjian (2014) note that Korean’s status as a classifier language may further complicate the analysis.

Bale & Khanjian (2008) show in (6) that marked nouns have an exclusive interpretation in most downward entailing contexts as *bezdig-ner* is interpreted as ‘two or more’ in the three contexts of a question, a conditional, and when under the scope of a universal quantifier.

- (6)
- a. *bezdig-ner uni-s?*
child-PL have-2SG
‘Do you have (two or more) children?’
 - b. *yete bezdig-ner uni-s, dun kena.*
if child-PL have-2SG, home go.2SG
‘If you have (two or more) children, then go home!’
 - c. *amen mart vor bezdig-ner uni-r vodk-i gajne-ts-av.*
all person that child-PL have-PST.3SG foot-DAT stand.up-PST-3SG
‘Everyone who had (two or more) children stood up.’

However, they do note that speakers preferred and tended to suggest an unmarked noun when eliciting these sentences. Furthermore, Bale & Khanjian (2008) also found that this exclusive interpretation of the marked plural is not available in negative sentences, a case where Western Armenian seems to pattern with English.

- (7) *bezdig-ner chi desah.*
child-PL NEG saw.1SG
‘I didn’t see children.’
 $\rightsquigarrow$ *I didn’t see one or more children*

They mention that speakers would find the sentence in (7) to be false if they saw one child, suggesting an inclusive interpretation. This contrasts with the other downward entailing contexts, which license only an exclusive interpretation of the plural. This pattern could make sense in the background of not all downward entailing contexts being equal and therefore permitting different interpretations of the plural. As Bale & Khanjian (2008) discuss, negation can be an unreliable downward entailing context. Additionally, questions are not always considered to be downward entailing (Guerzoni & Sharvit 2007), and universal quantifiers are downward entailing on the internal argument, but still upward entailing on their external argument (Crain 2017). While simply speculation, it could be plausible that an inequality of downward entailing contexts could result in an inequality of interpretations of plurals across contexts.

Concerning Turkish, Bale et al. (2011) show in (8), that the unmarked noun *çocuk* ‘boy’ can be used as a predicate for singular subjects as well as conjoined ones. But this is not the case for the *çocuklar* ‘boys’, which can only be a predicate for a conjoined subject. Görgülü (2012) and Bale et al. (2011) argue that Turkish’s *-lar* restricts the meaning of the noun from atoms and pluralities to only pluralities, which stands in contrast to English’s *-s*, which expands the meaning of the noun from only atoms to also include pluralities.

- (8)
- a. John çocuk.
John boy
‘John is a boy.’
 - b. John ve Brad çocuk.
John and Brad boy
‘John and Brad are boys.’
 - c. #John çocuk-lar.
John boy-PL
‘John is a boy.’
 - d. John ve Brad çocuk-lar.
John and Brad boy-PL
‘John and Brad are boys.’

However, recent experimental evidence shows that the Turkish marked plural is actually semantically aligned with English’s plural in that it is interpreted inclusively in negative phrases. Renans et al. (2020) ran two experiments designed to test the effects of monotonicity on the interpretation of the plural, as well as children’s acquisition of plural meaning in Turkish. Their results are in support of plural marked nouns having an inclusive interpretation in negative sentences. However, since only positive and negative sentence types were tested, it could be the case that Turkish and Western Armenian follow the same pattern of being exclusive in most downward entailing contexts, but just not under negation specifically.

The fact that these three languages function differently from English with respect to their plural semantics could be reasonably expected given their different morphological marking. Interestingly, Ahenkorah (2022) has claimed that Akan has an exclusive plural in downward entailing contexts, though its plural marking system differs from both English-type plural marking and Turkish-type plural marking in that both singular and plural nouns are explicitly marked.

2.1 Akan (Niger-Congo; Kwa)

As for a short description of Akan, it is a language in the Kwa subgroup of the Niger-Congo language family. It is the most widely spoken indigenous language in Ghana, with the last national census reporting around 45% of Ghana's population speaking it natively (about 7.7 million speakers), in addition to many more people speaking it as a second language. Most speakers identify with the specific variety of Akan that they speak, of which there are three major varieties: Ashante Twi, Akuapem Twi, and Fante. These are joined by multiple other smaller dialects of the language (Arkoh 2011). These varieties are generally mutually intelligible with some variation in pronunciation and vocabulary.

Concerning the basics of its grammar, Akan is an SVO language with a two-way tone system, and its syntactic categories are marked morphologically (Osam 2004).³ It is also said to previously have had a noun class system that is no longer productive, but has had lasting effects on the current grammar, such as in the number marking system. (Osam 1993; Appah 2003; Bodomo & Marfo 2006). However, the exact details of Akan's noun class system are not fully agreed upon in the literature. See Bodomo & Marfo (2006), cf. Osam (1993) for analyses of Akan's noun class system.

Akan's nominals are, in their most basic form, comprised of a root and a number-denoting prefix. The singulars, shown in (9a), take [ɔ-/o-], [ɛ-/e-] (modulo vowel harmony), or [a-] as their prefix. Some singular nouns have lost their prefix due to the attrition of the noun class system, and are marked with a null (∅) symbol in the examples. The plurals, in turn, are marked by a nasal ([n-] or [m-]) or an [a-] prefix. For both singular and plural nouns, the particular form of the prefix that appears depends on the noun class as well as some phonological processes. Unlike other noun class systems in the Niger-Congo family, these prefixes are not singular-plural pairs, though they may have been in a previous stage in the language's development.

- (9) a. ɔ-kraman e-fie o-yuo ∅-kaseɛ
 SG-dog SG-house SG-antelope SG-bone
- b. n-kraman a-fie a-yuo n-kaseɛ
 PL-dog PL-house PL-antelope PL-bone

³ Following the tradition of other academic works on Akan (Osam 1993; Appah 2003; Amo Ofori 2016; Ahenkorah 2022), tones are marked only for disambiguation when needed.

Noun class system attrition is also responsible for a handful of nouns in Akan lacking a singular-plural distinction, though they show remnants of the noun class system, as these nouns have retained a plural prefix.

- (10) a-kutuo n-homa n-yadoa
 NC-orange(s) NC-book(s) NC-garden.egg(s)

These nouns are typically regarded as number neutral or reported to be mass nouns by some speakers. While important to note the existence of these nouns, as the attrition of the noun class system is a change in progress, this research targets the semantic interpretation of the singular-plural distinction in Akan. Therefore, these words were therefore not included in any of the experimental items.

Most nominals in Akan use the strategy of noun class prefix alternation to mark number. However, certain subsets of nouns mark the plural with a circumfix, such as kinship terms. For these nouns, the singulars are still marked with only a prefix, just the way nouns in (9a) are, as shown in (11a). But the plural form of these nouns is then marked with a nasal or [a] as a prefix, along with *-nom* to complete the circumfix, shown in (11b).

- (11) a. o-nua a-sew ɔ-yiri
 SG-sibling SG-in-law SG-wife
 b. a-nua-nom n-sew-nom a-yiri-nom
 PL-sibling-PL PL-in-law-PL PL-wife-PL

A current analysis put forward for the semantics of Akan's plural prefix, as proposed by Ahenkorah (2022), states that it excludes atoms from its denotation, referring only to pluralities. In other words, the plural prefix is exclusive in both upward and downward entailing contexts. She provides (12) as evidence for her claims and explains that *nkonwa* 'chairs' can only refer to non-atomic entities. Therefore, the sentence in (12) means that Kofi did not buy a *plurality* of chairs; he could have bought no chairs or a singular chair, and the sentence would still be felicitous.

- (12) (Ahenkorah 2022: ex. 43a)
 Kofi a-n-tu n-konwa ɛnora.
 Kofi PERF-NEG-buy PL-chair yesterday
 'Kofi did not buy chairs yesterday.'

[The number of chairs bought by Kofi = 1 or 0]

This is contrasted with her analysis of the plural circumfix, where she describes this morpheme as having an inclusive interpretation in downward entailing contexts, just as the plural in English does. She provides the following example to illustrate:

(13) (Ahenkorah 2022: ex. 47)

Mary a-n-hu n-sew-nom wɔ paaki no so.
 Mary PERF-NEG-see PL-in.law-PL LOC park DET LOC
 ‘Mary didn’t see in-laws on the field.’

[The number of in-laws seen by Mary = 0]

Ahenkora’s example shows that *nsewnom* ‘in-laws’ includes the atomic reference, as Mary did not see either a plurality of in-laws or a singular in-law on the field.

Concerning the meaning contribution of each morpheme discussed, she states that the plural prefix takes away the atomic reference, resulting in nouns marked with just the prefix having an exclusive interpretation. She then describes the suffixal half of the circumfix as adding atomic reference back into the denotation, which gives the circumfixed plural its inclusive reading. She provides the denotation for *nsewnom* ‘in-laws’, which is shown in (14).

(14) (Ahenkorah 2022: pg. 28: ex.51)

$$\begin{aligned} \llbracket \text{-sew}' \rrbracket &= \{a, b, c, a+b, b+c, a+c, a+b+c\} \\ \llbracket \text{n-} \rrbracket &= \lambda P \lambda x [P(x) \wedge \neg AT(omic)(x)] \\ \llbracket \text{n-sew} \rrbracket &= \lambda P \lambda x [P(x) \wedge \neg AT(omic)(x)](sew') \\ \llbracket \text{n-sew} \rrbracket &= \{a+b, b+c, a+c, a+b+c\} \\ \llbracket \text{n-sew-nom} \rrbracket &= \lambda P \lambda x \exists y [P(y) \wedge x \leq y](n-sew) \\ &= \lambda x \exists y [(n-sew)(y) \wedge x \leq y] \\ &= \{a, b, c, a+b, b+c, a+c, a+b+c\} \end{aligned}$$

Here, *n-* removes the atoms from the derivation. If a noun were to stop at this stage, according to Ahenkorah (2022), it would have an exclusive interpretation in downward (and upward) entailing contexts. However, if a noun also has the *-nom* suffix (which is restricted to only kinship terms), then the noun is able to have an inclusive interpretation in downward entailing contexts, as the atomic referents are now included once more. That is to say, the circumfix, when treated as a whole, does not add or restrict the denotation of the noun, with *-nom* signaling kinship. It should be noted, however, that for some speakers, *-nom* is optional. In other words, some speakers would simply say *nsew* to mean ‘in-laws’, rather than *nsewnom*. It would then be predicted that *nsew* would have an exclusive interpretation in downward entailing contexts according to the analysis put forward by Ahenkorah (2022).

Akan is interesting to investigate in this respect since the current claims concerning the interpretation of the plural are set apart from Western Armenian, Turkish, and Korean not only in the fact that its plural marking system explicitly marks both singular and plural nouns, but also in that the proposed analysis by Ahenkorah (2022) suggests that only one of the plural markers yields only an exclusive interpretation in downward entailing contexts rather than having a unified

analysis for the plural nouns in the language. Additionally, the plural circumfix being essentially semantically null with respect to number is another reason this analysis should be questioned.

3 Hypotheses and predictions

The experiments described below are designed to evaluate Ahenkora's claims and provide a deeper look into Akan's plural semantics. More specifically, I ask if the inclusive interpretation of the prefixed and circumfixed plurals is available to Akan speakers in the downward entailing contexts of negation, in questions, and under the restrictor of universal quantifiers.

To answer this question, two nearly identical experiments were run. The first experiment concerns the plural prefix, and the second experiment concerns the plural circumfix. For each of the two experiments, three hypotheses are posited: The Exclusive Hypothesis, the Inclusive Hypothesis, and the Mixed Hypothesis. These hypotheses concern only the downward entailing contexts, as plurals are expected to only have an exclusive interpretation available in upward entailing contexts in all three hypotheses. Any of the three hypotheses described below are possible for each of the two experiments. That is to say, it is possible for the two plural morphemes tested to have different semantic interpretations, which should be the case, according to Ahenkora's (2022) analysis.

The first hypothesis, presented in (15) is the Exclusive Hypothesis, which takes inspiration from the claims made by Bale & Khanjian (2008; 2014); Görgülü (2012), and Kang (1994), as well as from Ahenkora's treatment of the plural prefix. This hypothesis is inline with Ahenkora's analysis for the prefixed plural, but is contrary to her analysis for the circumfixed plural. It predicts that speakers will only be able to access the exclusive interpretation of the plural in downward entailing contexts.

- (15) **Exclusive Hypothesis:** The exclusive reading is the only available interpretation in downward entailing contexts for prefixed / circumfixed plurals in Akan.

An exclusive reading of the plural is characterized by a low score (< 2) for the [Positive] and [Question] sentence types and a high score (> 4) for [Negative] and [Universal Quantifier sentence types], as shown in **Figure 1**. The expectations of each hypothesis for each of the sentence types are described in detail in Sections 3.1–3.4. In the statistical analysis, this hypothesis would predict a main effect of sentence type, as the level of acceptability is expected to vary across sentence type. However, since this hypothesis does not predict the plural to differentiate semantically between one and zero references, it predicts no main effect of context. Given that the difference of slope between the two contexts doesn't change, as seen in (1), the Exclusive Hypothesis also does not predict a significant interaction between the the factors of sentence type and context.

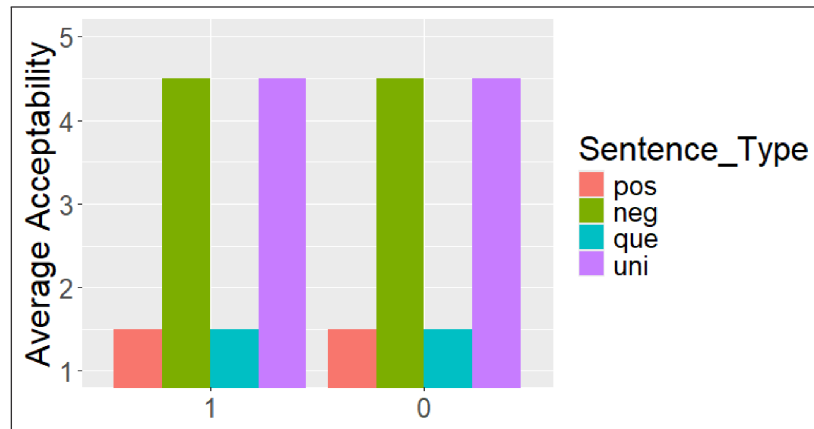


Figure 1: The predictions made by the Exclusive Hypothesis for Experiment 1 & 2.

The second hypothesis put forward for this research, the Inclusive Hypothesis, contrasts with Ahenkora's claim for the plural prefix in Akan, though it is in line with her analysis for the plural circumfix. It predicts that plurals in Akan function closer to how English's plurals do. In other words, it predicts that plurals include the atomic denotation in the selected downward entailing contexts, and that the inclusive interpretation is available to speakers.

- (16) **Inclusive Hypothesis:** Both inclusive and exclusive interpretations of the plural are available in downward entailing contexts for prefixed / circumfixed plurals in Akan.

A result that would indicate that plurals in Akan are inclusive in downward entailing contexts would be characterized by each of the downward entailing sentence types ([Negative], [Question], and [Universal Quantifier]) in context [1] having an intermediate average rating (between 2 and 4), as displayed in **Figure 2**. This hypothesis also predicts a main effect of sentence type. However, in a departure from the Exclusive Hypothesis, the Inclusive Hypothesis predicts a significant interaction between sentence type and context, as the availability of the exclusive reading is expected to vary based on not only context, but also on the entailment pattern of the sentence type. This interaction is characterized by significant interactions for all three downward entailing sentence types.

Lastly, there is a third hypothesis to take into consideration. According to data presented by Bale & Khanjian (2008), the plural in Western Armenian has an exclusive interpretation in downward entailing contexts, *except* for in the context of negation, resulting in Western Armenian's plural falling into a 'mix' of the exclusive and inclusive hypotheses, as described in Section 2. While their discussion of this phenomenon is not explicitly a prediction of an additional type of language, the pattern reported in Bale & Khanjian (2008) informs the third hypothesis for the two experiments. The Mixed Hypothesis is as follows:

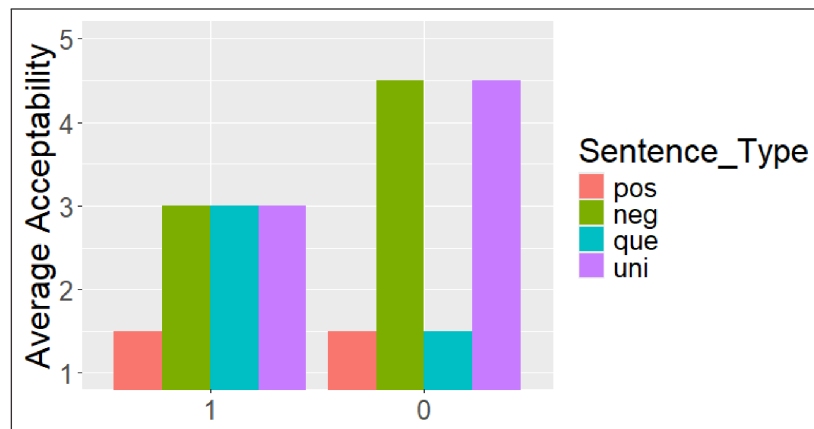


Figure 2: The predictions made by the Inclusive Hypothesis for Experiment 1 & 2.

- (17) **Mixed Hypothesis:** The exclusive reading is the only available interpretation for prefixed / circumfixed plurals in questions and when under the scope of a universal quantifier. Both inclusive and exclusive interpretations of the plural are available in negative sentences for prefixed / circumfixed plurals in Akan.

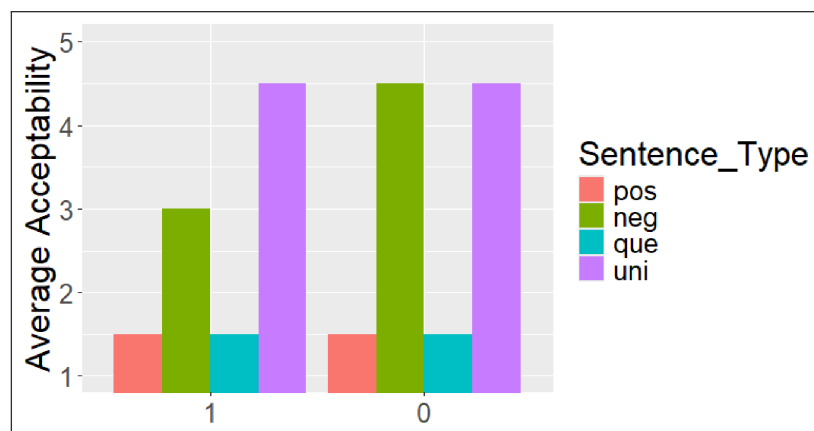


Figure 3: The predictions made by the Mixed Hypothesis for Experiment 1 & 2.

Under this hypothesis, nearly the same outcome as the Exclusive Hypothesis is expected. However, the difference lies in the prediction for negation. The Mixed Hypothesis would find the [Negative] sentence type to follow the predictions of the Inclusive Hypothesis, i.e., the inclusive interpretation of the plural should be available in the negative sentences for speakers and it would have an intermediate rating between 2 and 4, as displayed in **Figure 3**. Meanwhile, the inclusive interpretation should not be available in the other two downward entailing contexts, resulting in a low (< 2) and high (> 4) average for the [Question] and [Universal Quantifier] sentence types,

respectively. The Mixed Hypothesis predicts a main effect of sentence type and an interaction of the factors sentence type and context. However, this interaction differs from the interaction predicted by the Inclusive Hypothesis. Under the Mixed Hypothesis, only the [Negative] sentence type should display a significant interaction.

For each hypothesis, the predictions for each of the four sentence types tested in the experiments ([Positive], [Negative], [Question], [Universal Quantifier]) are provided with example items below. The impact of sentence type on context [1] is particularly of interest, as the predictions that each hypothesis makes for the [0] context (a context in which none of the objects are acted on) are stable, allowing it to serve as a control context. The [1] and [0] contexts delineate the number of objects that are acted upon in the context sentences. The structure of the example sentences shows both possible contexts for ease of presenting the predictions that the hypotheses make. However, during the experiments, only one context per item was presented to the participant, followed by the sentence type being tested (see Section 4.1.1 for a detailed description of the experimental procedure).

In the following section, I discuss the specifics of the predictions each hypothesis makes for each of the experimental conditions by way of a sample item the participant would have seen during the experiment.

3.1 Sentence type: Positive

- (18) **CONTEXT [1]** : Aborosan bi ben Kojo fie. Mpen pii no, mmarima kakraa bi na wɔreyɛ adwuma wɔ hɔ. Nanso ɛnnɛ, Kojo hunuu ɔbarima baako se ɔreyɛ adwuma.
 ‘There’s a storey building near Kojo’s house. Usually, there’s a few men working there. But today, Kojo only saw one man working.’
CONTEXT [0] : Aborosan bi ben Kojo fie. Mpen pii no, mmarima kakraa bi na wɔreyɛ adwuma wɔ hɔ. Nanso ɛnnɛ, Kojo anhunu mmarima biara a wɔreyɛ adwuma.
 ‘There’s a storey building near Kojo’s house. Usually, there’s a few men working there. But today, Kojo didn’t see any men working.’
SENTENCE TYPE [POSITIVE] :
 Kojo hunu-u m-marima.
 Kojo see-PST PL-man
 ‘Kojo saw men.’

According to the Exclusive Hypothesis, in the example item shown here in (18), the test sentence ‘Kojo saw men,’ should be understood as Kojo having seen more than one man. In this view, the sentence would be infelicitous in both contexts that describe Kojo as having seen either one man or zero men. The Inclusive Hypothesis would similarly interpret the plural *mmarima* ‘men’ as exclusive, as this hypothesis predicts that the inclusive reading is available only in downward entailing contexts. Lastly, the Mixed Hypothesis also predicts that *mmarima* ‘men’ would be

interpreted exclusively in positive sentences. All three hypotheses predict a low average score (< 2) for positive sentences in both [0] and [1] contexts.

3.2 Sentence type: Negative

- (19) **CONTEXT [1]** : Nana nsewnom nyinaa gye n'asebaa pe watoa nananom wo nseedo, enti wo asew baako.
 'All of Nana's in-laws except for his mother-in-law have passed away, so he only has one in-law.'
CONTEXT [0] : Nana nwaree, enti onni nsewnom.
 'Nana is not married, so he has no in-laws.'
SENTENCE TYPE [NEGATIVE] :
 Nana n-ni n-sew-nom.
 Nana NEG-have PL-in.laws-PL
 'Nana doesn't have in-laws.'

Under the Exclusive Hypothesis, the negative test sentence in (19), *Nana nni nsewnom*. 'Nana doesn't have in-laws' would be interpreted as Nana not having more than one in-law. In other words, it's possible for this sentence to still be true if he has either zero in-laws or one in-law. Therefore, the [Negative] sentence type should be considered acceptable in both [0] and [1] contexts under the Exclusive Hypothesis, and therefore, receive a high average score (> 4). As for the Inclusive Hypothesis, it would interpret *nsewnom* as meaning 'one or more in-laws' in the downward entailing context of negation. Therefore, this hypothesis predicts that participants would give this test sentence a high average rating in context [0]. If only the inclusive reading of the plural is accessed, participants would rate this test item quite high in context [1]. However, the Inclusive Hypothesis predicts that both inclusive and exclusive readings are available. On account of previous similar experimental studies such as Renans et al. (2020) and Tieu et al. (2020), it would be expected that the full range of ratings are selected by participants, averaging out to an intermediate score near the middle of the scale if both readings are accessed by participants. The Mixed Hypothesis is informed by Bale & Khanjian's (2008) analysis of Western Armenian, which does have an inclusive interpretation of the plural in negative sentences. So for this sentence type, the Mixed Hypothesis predicts the same judgments as the Inclusive Hypothesis: a high average score (> 4) for context [0], and an intermediate average score ($> 2, < 4$) for context [1].

3.3 Sentence type: Question

- (20) **CONTEXT [1]** : Amma ne Abena reka wo abusua ho asm. Abena babaa, Afia aware mfee kakraa bi na wo ba baako.
 'Amma and Abena are talking about their families. Abena's daughter, Afia has been married for a few years, and has one child.'

CONTEXT [0]: Amma ne Abena reka wɔn abusua ho asem. Abena babaa, Afia waree nnasa yi, nanso ɔnnya nwoee.

‘Amma and Abena are talking about their families. Abena’s daughter, Afia has recently gotten married, but does not have children yet.’

SENTENCE TYPE [QUESTION]:

Amma bisa Abena: “Afia wɔ m-ma?” Abena bua-a se “Aane.”

Amma ask.3SG Abena: Afia have.3SG PL-child? Abena say.3SG-PST as yes

‘Amma asks Abena: Does Afia have children? Abena answers “Yes”.’

For the [Question] sentence type, the Exclusive Hypothesis would interpret *mma* ‘children’ to mean ‘more than one child’, and would therefore see the test sentence to be infelicitous in both context [0] and context [1], receiving a low average score (< 2). The Inclusive Hypothesis predicts that *mma* would be interpreted as ‘one or more children’, and the sentence would then be rated with an intermediate average score ($> 2, < 4$) in context [1], as both inclusive and exclusive readings are predicted to be available and a low average score (< 2) in context [0]. For the [Question] sentence type, the Mixed Hypothesis patterns with the Exclusive Hypothesis and thus also predicts that the test sentence would be scored quite low in both contexts.

One may expect the predictions of the acceptability judgments for each of the downward entailing contexts to be in alignment. However, due to the formulation of the [Question] sentence type, the hypotheses predict the opposite acceptability judgment than what they predict for the [Negative] and [Universal Quantifier] sentence types. It must be stated clearly that while the predictions for the average scores of acceptability judgments differ, the predictions for the interpretation of the plural do not. In each downward entailing context, the Exclusive Hypothesis predicts for it to be interpreted as ‘more than one’, while the Inclusive Hypothesis predicts that it will be interpreted as ‘one or more’ with the dispreferred reading being accessible. The Mixed Hypothesis, which follows the Bale & Khanjian (2008) analysis for Western Armenian, predicts that plurals in questions would be interpreted exclusively.

3.4 Sentence type: Universal quantifier

(21) **CONTEXT [1]:** ɛse se Kofi ne n’ayɔnkofoɔ bisa nananom mmiensa nsem a a efa wɔn abrabɔ ho. Ne yɔnko biara bisaa nananom mmienu anaa deɛ ɛboro saa. Nsem a ɛfiri nananom ye, ne n’ayɔnkofoɔ tumi hye ahyee twere wɔn amanneɛ. ɛse se Kofi ne n’ayɔnkofoɔ bisa na ɔntumi nhye asee ntwere amanneɛ no.

‘Kofi and his classmates must interview grandparents about their life. Each of his classmates were able to interview two or more grandparents. With enough information from the grandparents, his classmates can now begin writing the report. Kofi was only able to interview one grandparent and cannot start writing the report.’

CONTEXT [0]: ɛsɛ sɛ Kofi ne n'ayɔŋkofoɔ bisa nananom mmiensa nsem a a ɛfa wɔn abrabɔ ho. Ne yɔŋko biara bisaa nananom mmienu anaa deɛ ɛboro saa. Nsɛm a ɛfiri nananom yɛ, ne n'ayɔŋkofoɔ tumi hyɛ ahyɛ twɛrɛ wɔn amanneɛ. Kofi antumi ammisaa nananom biara asem, na ɔntumi nhyɛ aseɛ ntwɛrɛ amanneɛ no.

'Kofi and his classmates must interview grandparents about their life. Each of his classmates were able to interview two or more grandparents. With enough information from the grandparents, his classmates can now begin writing the report. Kofi wasn't able to interview any grandparents, and cannot start writing the report.'

SENTENCE TYPE [UNIVERSAL QUANTIFIER]:

Obiara a ɔ-bisa-a n-nana-nom n-sem tumi hyɛ aseɛ twɛrɛ
everyone REL 3SG-ask-PST PL-grandparent-PL PL-word MOD put under write
amanneɛ no.

report DEF

'Everyone who interviewed grandparents can begin writing the report.'

Under the scope of a universal quantifier, the Exclusive Hypothesis predicts that participants interpret *nnananom* 'grandparents' as a strict plurality: 'more than one grandparent'. This hypothesis leads us to expect high average ratings for both [0] and [1] contexts. The Inclusive Hypothesis, however, would interpret *nnananom* as 'one or more grandparents', which would result in an intermediate average rating (> 2 , < 4) for context [1] and a high average rating (> 4) for context [0]. The predictions of the Mixed Hypothesis follow those of the Exclusive Hypotheses and predict a high average score (> 4) for both contexts as well.

4 The experiments

To determine which of the above hypotheses describe the actual situation for Akan, two experiments were run simultaneously. One experiment concerned nouns with the plural prefix (regular plurals), and the other, the plural circumfix (kinship terms). These experiments were designed to be nearly identical to allow for them to be as comparable as possible.

Participants could elect to take both experiments or only one. To avoid the ordering of the experiments possibly influencing the outcome or priming participants, each participant was assigned to one of the experiments at random. To illustrate, some participants completed Experiment 1 first, some completed Experiment 2 first. All of Experiment 1 participants also did Experiment 2. However, two participants who completed Experiment 2 first elected to not do Experiment 1, resulting in a difference in the number of participants for each experiment. To account for this, a random intercept for Participant is included in the model.⁴ The relatively low number of participants in both experiments is a result of the difficulty in finding native Akan

⁴ Unfortunately, a model with a random slope for Participant did not converge.

speakers who have the desire and ability to take part in the experiment. Furthermore, similar experiments conducted previously also had a similar number of participants per experiment, at least for the children. Renans et al. (2017) tested 45 adults and 21 children and Tieu et al. (2020) tested 43 adults and 28 children in Experiment 1, 27 adults and 17 children in Experiment 2, and 24 adults in Experiment 3.

Concerning the statistical analysis, the results of the experiments were analyzed using the same methods: both analyses were done using a linear mixed effects analysis performed using `lmer4` (Bates et al. 2015) in the R environment (R Core Team 2023). The model contained the fixed effects of `CONTEXT` and `SENTENCE TYPE`, as well as random intercepts for `ITEM` and `PARTICIPANT`. *P*-values and χ^2 s for the main effects were obtained through likelihood ratio tests of the full model against the model without the target effect, and the *p*-value and χ^2 for the interaction was obtained through a likelihood ratio test of the full model with an interaction term against the full model without an interaction term. Anonymized data sets and scripts used for analysis can be found at https://osf.io/3n46m/?view_only=183cafd9e316442f974f8b2d741a66cf.

4.1 Experiment 1: Plural prefix

The first experiment conducted was designed to test the effect of monotonicity on Akan's prefixed plurals. It seeks to answer the question: in downward entailing contexts, can participants access the inclusive interpretation of the plural marked with the plural prefix?

4.1.1 Methods

In this experiment, the two-level factor `CONTEXT` [0, 1] was fully crossed with the four-level factor `SENTENCE TYPE` [Positive, Negative, Question, Universal Quantifier], providing eight total experimental conditions [Positive-0, Negative-0, Question-0, Universal Quantifier-0, Positive-1, Negative-1, Question-1, Universal Quantifier-1]. Using a Latin Square design for this within-subjects experiment, each participant heard three test items for each of the eight conditions, resulting in 24 test items having been presented in a random order to participants. Furthermore, 12 filler items were also interspersed randomly between the test items. Each item was recorded by a native speaker and presented to the participants in an auditory fashion. This method of presentation was chosen since, despite having a standard orthographic system, Akan is primarily a spoken language, with the majority of writing in Ghana being in English.

4.1.2 Participants

For this experiment, 21 native Akan speakers were recruited at the University of Ghana, as well as through personal networks. Participants were selected if they were over 18 years old, self-identified as being a native speaker of any variety of Akan, and if their parents also spoke a variety

of Akan. Each participant was awarded the equivalent of five euros for their participation in each experiment. Two participants were excluded from the analysis due to selecting high ratings for all items, indicating an unreliability of the data from these two participants, resulting in data from 19 participants being analyzed for Experiment 1.

4.1.3 Procedure

In the experiment, participants first heard the context sentence (labeled as CONTEXT in the examples in Section 3), then a test sentence (labeled as SENTENCE TYPE). After hearing both context and test audios play all the way through, the participant was asked to judge how acceptable each test sentence would be to say in the given context on a five-point scale, with 1 being the worst and 5 being the best.⁵

4.1.4 Materials

The test items for both experiments were created with the intention of directly comparing the two experiments. Because the plural circumfix (Experiment 2) is limited to only kinship nouns, the items for Experiment 1, which investigated the plural prefix, were limited to only using human nouns as the target plural. This was done to avoid any effects of animacy on the interpretation of the plural, therefore. Items were designed with the help of native Akan speakers. They were also proofread by an additional speaker to avoid any errors. The items outlined in Sections 3.1 through 3.4 are real test items that were presented to participants. Items in (18) and (20) were used in Experiment 1, as they have prefixed plurals as the target vocabulary: *mmarima* ‘men’ and *mma* ‘children’, respectively.

4.1.5 Results

Experiment 1 resulted in 456 observations of 24 test items from 19 participants. **Figure 4** shows the average acceptability for each condition, followed by a table summarizing the acceptability judgments.

A linear mixed-effects model was fitted to the acceptability scores given by participants, with CONTEXT, SENTENCE TYPE, and their interaction as fixed effects, and PARTICIPANT and ITEM as random intercepts. The mean and standard deviations are reported in **Table 1**. The model coefficients are reported in **Table 2**. A comparison between the full model and a reduced model without the interaction term showed that the inclusion of the interaction significantly improved the model fit (likelihood-ratio test: $\chi^2(7) = 45.214$, $p < 0.001$).

⁵ The use of a Likert scale was chosen, against the advice of Bombi & Veagh-Geiss (2018), as it is necessary to allow participants to select a score which reflects that they find the test sentence mildly acceptable without fully committing to finding it maximally acceptable, as they would need to in a forced-choice task.

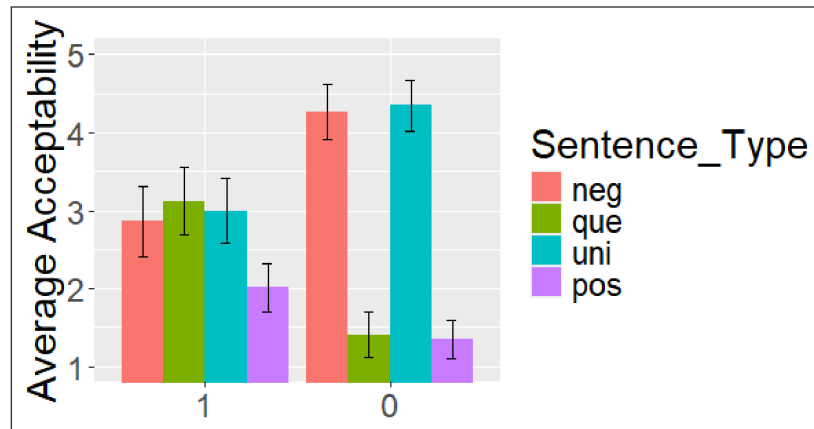


Figure 4: A bar graph of the results from Experiment 1.

Context	Sentence_Type	Mean_judge	SD_judge	SE_judge
1	pos	2.018	1.188	0.1573
1	neg	2.86	1.737	0.23
1	que	3.123	1.659	0.2197
1	uni	3.0	1.57	0.2079
0	pos	1.351	0.9352	0.1239
0	neg	4.263	1.357	0.1797
0	que	1.404	1.116	0.1478
0	uni	4.351	1.261	0.167

Table 1: The average acceptability ratings from Experiment 1.

Term	Est. (β)	SE	<i>t</i> value	df	Lower CI	Upper CI	<i>p</i> value
(Intercept)	2.791	0.125	22.32	41.17	2.54	3.042	<0.001
Sentence_Type_Neg	0.681	0.206	3.304	51.23	0.265	1.089	0.002
cont Sentence_Type_Que	-0.473	0.206	-2.295	51.23	-0.882	-0.058	0.026
Sentence_Type_Uni	0.807	0.206	3.916	51.23	0.391	1.216	<0.001
Context_1	0.069	0.119	0.578	51.23	-0.168	-0.168	0.566
Sentence_Type_Neg:Context_1	0.739	0.206	3.585	51.23	0.331	1.155	<0.001
Sentence_Type_Que:Context_1	-0.960	0.206	-4.66	51.23	-1.375	-0.551	<0.001
Sentence_Type_Uni:Context_1	-0.681	0.206	-3.302	51.23	-1.095	-0.271	0.002

Table 2: β -coefficients for Experiment 1.

4.1.6 Discussion

Both conditions for the [Positive] sentence type were found to be generally unacceptable by participants, as predicted by each of the three hypotheses. The results seen in both the graph and the statistics of Experiment 1 best support the Inclusive Hypothesis. The average rating of near 3 for all downward entailing contexts paired with the range of judgments, as indicated by the standard deviation shows that the inclusive reading was accessed alongside the exclusive reading. A main effect of Sentence Type and an interaction between the two factors, characterized by a significant interaction for all three downward entailing sentence types is in line with the Inclusive Hypothesis. This therefore indicates that, contrary to Ahenkora's analysis of the plural prefix, Akan speakers can interpret the prefixed plural inclusively, in a similar manner to how English speakers interpret the English plural. A discussion of these results in conjunction with the results from Experiment 2 is continued in Section 5.

4.2 Experiment 2: Plural circumfix

The second experiment in this study was designed to test the plural circumfix in Akan. The question remains as to whether speakers can access the inclusive interpretation of circumfixed plurals in downward entailing contexts, as claimed by Ahenkorah (2022), or if only the exclusive interpretation is available.

4.2.1 Methods

The methods and design of Experiment 2 mirror those of Experiment 1 as closely as possible for the purpose of direct comparison between the two experiments. The key difference concerns only the items, which use kinship terms (nouns which take the *n-/m-/a-...nom* plural circumfix in Akan), rather than the typical nouns which take only the plural prefix.

4.2.2 Participants

For this experiment, 23 participants were recruited through personal networks and at the University of Ghana. The two experiments were always advertised as a pair. Each participant was awarded five euros for their participation in each experiment. Those who participated in both experiments would receive a total of ten euros. Two participants were excluded from the analysis due to selecting high ratings for all items, indicating an unreliability of the data from these two participants, resulting in data from 21 participants being analyzed for Experiment 2.

4.2.3 Procedure

The procedure of the experiment was exactly the same as that of Experiment 1: The participant first heard the context, followed by the test sentence, then was asked to judge the test sentence on a five-point scale. At the end of the experiment, the participant had the opportunity to leave comments. For further details on the procedure, refer back to Section 4.1.3.

4.2.4 Materials

The target plural of each item was a kinship term, since the *n-/m-/a-...nom* plural marking strategy is restricted to this domain of vocabulary. (19) and (21) from Section 3 have kinship as target vocabulary: *nsewnom* ‘in-laws’ and *nnananom* ‘grandparents’, respectively.

4.2.5 Results

The method of analysis for Experiment 2 was exactly the same as that of Experiment 1. Refer to Section 4.1.5 for a detailed description. Experiment 2 resulted in 504 observations of 24 test items from 21 participants. **Figure 5** shows the average acceptability for each condition, followed by a table summarizing the acceptability judgments.

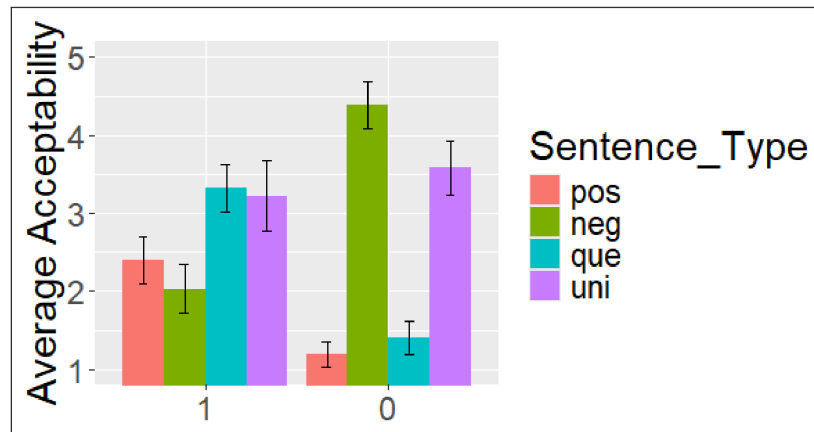


Figure 5: A bar graph of the results from Experiment 2.

Context	Sentence_Type	Mean_judge	SD_judge	SE_judge
1	pos	2.397	1.238	0.156
1	neg	2.032	1.244	0.1567
1	que	3.317	1.216	0.1532
1	uni	3.222	1.521	0.2267
0	pos	1.19	0.6686	0.0842
0	neg	4.381	1.211	0.1525
0	que	1.397	0.8714	0.1098
0	uni	3.58	1.564	0.1738

Table 3: The average acceptability ratings from Experiment 2.

Term	Est. (β)	SE	<i>t</i> value	df	Lower CI	Upper CI	<i>p</i> value
(Intercept)	2.665	0.1003	26.57	35.09	2.463	2.869	< 0.001
Sentence_Type_Neg	0.5433	0.1576	3.448	40.92	0.23	0.862	0.001
Sentence_Type_Que	-0.2587	0.1576	-1.642	40.92	-0.570	0.064	0.108
Sentence_Type_Uni	0.58	0.1622	3.576	40.83	0.241	0.899	< 0.001
Context_1	0.03657	0.09168	0.3988	40.91	-0.148	0.22	0.69
Sentence_Type_Neg:Context_1	-1.144	0.1576	-7.256	-1.455	-0.820	-0.264	< 0.001
Sentence_Type_Que:Context_1	0.8655	0.1576	5.492	40.92	0.545	1.179	< 0.001
Sentence_Type_Uni:Context_1	-0.3012	0.1623	-1.855	-0.632	0.021	-0.542	0.071

Table 4: β -coefficients for Experiment 2.

A linear mixed-effects model was fitted to the acceptability scores given by participants, with CONTEXT, SENTENCE TYPE, and their interaction as fixed effects, and PARTICIPANT and ITEM as random intercepts. The mean and standard deviations are reported in **Table 3**. The model coefficients are reported in **Table 4**. A comparison between the full model and a reduced model without the interaction term showed that the inclusion of the interaction significantly improved the model fit (likelihood-ratio test: $\chi^2(7) = 42.284$, $p < 0.001$).

4.2.6 Discussion

The results from Experiment 2 suggest that both inclusive and exclusive interpretations are available for the plural circumfix in downward entailing contexts, given the intermediate means for each of the downward entailing sentence types for context [1]. The standard deviations also suggest participants had a range of interpretations that resulted in these averages, similarly to Experiment 1, although the standard deviations in Experiment 2 are a bit smaller. Interestingly, the [Negative] sentence type was rated lower in Experiment 2 than it was in Experiment 1, with an average acceptability rating of only 2.032. This could suggest a more strict inclusive interpretation for circumfixed plurals, or in other words, a lesser availability of the exclusive reading in negative sentences for the plural circumfix. The statistics show a main effect of sentence type and a significant interaction. However, the interaction of the [Universal Quantifier sentence type] was barely significant. Additionally, the [Universal Quantifier-1] and [Universal Quantifier-0] conditions have very similar average ratings (3.222 and 3.58, respectively) from participants. The intermediate average rating for [Universal Quantifier-1] falls into the pattern predicted by the Inclusive Hypothesis, though the intermediate average rating for [Universal Quantifier-0] is not predicted by any of the three hypotheses. This suggests that participants likely interpret kinship plurals as inclusive under the scope of a universal quantifier, but that there may have been issues with the items themselves. The lower average rating for [Universal Quantifier-0] could be a result of the design of the [Universal Quantifier] sentence types, as the recording of

the context description took more time to play through than the other three sentence types, which were much shorter in comparison. Perhaps participants became distracted or paid attention to a part of the context that was not directly connected of the number of people in the situation for each item due to the lengthier descriptions, resulting in a slightly lower than expected average rating for the [Universal Quantifier-0] condition. Despite this, the results seem to best align with the Inclusive Hypothesis, which is in line with Ahenkora's (2022) description of this morpheme.

5 General discussion

Although these experiments have the same overall conclusion, there are some slight differences in between the two. Interestingly, participants showed a stronger preference for the inclusive interpretation for negative items in context [1] in Experiment 2 than they did in Experiment 1. Participants also showed less variation (a lower standard deviation) in their scores in Experiment 2 than they did in Experiment 1, suggesting a slightly more uniform interpretation of the circumfixed plural within the community. These differences might be attributed to a variety of factors such as length of items (as discussed in Section 4.2.6 for the [Universal Quantifier] sentence type) or simply the lower than ideal number of participants. Additionally, the [Positive] sentence type in context [1] was rated higher than expected by any of the three hypotheses. While not typical for human nouns, Akan does have some nouns which lack a singular-plural distinction, as mentioned in example (10). For a language which has experienced considerable and potentially ongoing attrition of its noun class system, on which the singular-plural distinction hangs, it might not be so odd to find that the inclusive interpretation of plurals is moderately available even in positive sentences.

Ultimately, however, the results of both experiments show that Akan's plural, no matter the marker, has an inclusive interpretation in all three downward entailing contexts tested. The results of these two experiments are in line with Ahenkora's analysis of the circumfixed plural, but are not in agreement with her proposal for the plural prefix. These results, therefore, invite a reevaluation of the semantic derivations for each morpheme proposed by Ahenkorah (2022).

5.1 Reassessing the plural circumfix

Ahenkorah (2022) proposed that the prefixed plural is exclusive, and the circumfixed plural is inclusive. She considers the root to be number neutral and for the prefix and the prefix half of the circumfix to remove the atomic reading from the noun. However, to get the inclusive interpretation of the circumfixed plural, she proposes that the suffix part of the circumfix adds the atoms back into the derivation. Her derivation of the noun *nsewnom* 'in-laws', which was introduced in Section 2.1 shows this, and is reproduced here, in (22).

- (22) (Ahenkorah 2022: pg. 28: ex.51)
- $$\begin{aligned} \llbracket \text{-sew}' \rrbracket &= \{a, b, c, a+b, b+c, a+c, a+b+c\} \\ \llbracket \text{n-} \rrbracket &= \lambda P \lambda x [P(x) \wedge \neg AT(omic)(x)] \\ \llbracket \text{n-sew} \rrbracket &= \lambda P \lambda x [P(x) \wedge \neg AT(omic)(x)](sew') \\ \llbracket \text{n-sew} \rrbracket &= \{a+b, b+c, a+c, a+b+c\} \\ \llbracket \text{n-sew-nom} \rrbracket &= \lambda P \lambda x \exists y [P(y) \wedge x \leq y](n-sew) \\ &= \lambda x \exists y [(n-sew)(y) \wedge x \leq y] \\ &= \{a, b, c, a+b, b+c, a+c, a+b+c\} \end{aligned}$$

However, her analysis cannot account for the results of the two experiments presented above, where both types of plurals have an inclusive interpretation. I propose that these kinship terms and regular nouns are both marked for number with just the plural prefix and that the *-nom* suffix is simply a kinship marker. This would then change the status of the plural circumfix to two elements: a plural prefix and a kinship suffix.

This division of the circumfix is further supported by the fact that some speakers find the *-nom* part of the circumfix to be optional. Ahenkorah (2022) herself shows that the plural interpretation of ‘in-laws’ can still be understood without it, as (23b) *nsew* is still felicitously interpreted as ‘in-laws’ despite missing *-nom*. However, it is not possible to obtain a plural interpretation without the prefix. (23d) shows *sewnom* as an ungrammatical form for the intended interpretation of ‘in-laws’.

- (23) a. a-sew ‘in-law’
 b. n-sew ‘in-laws’
 c. n-sew-nom ‘in-laws’
 d. *sew-nom ‘in-laws’

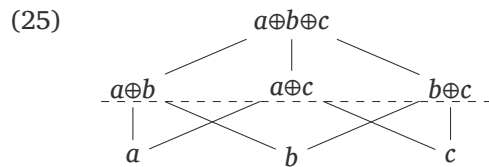
An additional alleged plural circumfix does exist in Akan: *n-/m-/a-...foɔ*, which was not included in the above experiments, though Ahenkora’s analysis concerning the effect of monotonicity on *n-/m-/a-...foɔ* matches her analysis of *n-/m-/a-...nom*. Both *n-/m-/a-...foɔ* and *n-/m-/a-...nom* occur with a restricted domain of nouns, both only applying to a subset of human nouns. In the case of *n-/m-/a-...foɔ*, these morphemes are described by Appah (2003) to be a nominalization strategy where the resulting noun “refers to the agent or actor in the action or process designated by the verb” (Appah 2003: 87), two of which are shown in (24). However, this morpheme, unlike *n-/m-/a-...nom*, has a counterpart, *V-...-ni*, which marks the singular form.

- (24) a. o-sika-ni a-sika-foɔ
 SG-money-NI PL-money-FOO
 ‘rich person’, ‘rich people’

- b. ɔ-sukuu-ni a-sukuu-foɔ
 SG-school-NI PL-school-FOɔ
 ‘student’, ‘students’

Dolphyne (1988) states that *-ni* and *-foɔ* developed from *oni* ‘person’ and became a nominalization strategy as a suffix. It may be of interest to look into the historical development of the singular counterparts of kinship terms, for which there’s currently no diachronic analysis. It could well be the case that singular kinship terms had previously been marked with a singular counterpart to *-nom*, that has over time become unpronounced. This would be in line with the pattern of attrition seen in the noun class system, where most of the attrition seems to occur on singular nominals, as well as in the adjectival agreement with singular nouns (Osam 1993). Morphemes which co-occur do not necessarily indicate identical meaning contributions. This is to say that the suffixal half of these plural circumfixes contribute [+kinship] in the case of *n-/m-/a-...nom* or [+agent] in the case of *n-/m-/a-...foɔ*. This leaves the plural/singular prefix to do the heavy lifting of number marking.

Returning to the meaning of the prefixes in Akan, a unified analysis for both sets of words tested in Experiments 1 and 2 is in order. I suggest that nouns with a singular prefix denote only atoms, i.e., only the part of the semi-lattice below the dotted line in (25). Nouns marked with a plural prefix then denote the whole semi-lattice: atoms and pluralities.



In summary, given that the experimental evidence presented shows that both types of plurals tested have an inclusive interpretation, the only plural-denoting prefix in Akan is the plural prefix. The morphemes previously analyzed as plural circumfixes are instead a combination of the plural prefix and a kinship- or agent-denoting suffix.

6 Conclusion

The present study examined the effect of monotonicity on two plural morphemes in Akan in light of claims of an exclusive plural in both upward and downward entailing contexts. To test if the plural prefix and/or the plural circumfix have an inclusive or exclusive reading, two experiments containing multiple downward entailing contexts were run to obtain native speaker judgments. The results indicate that both morphemes tested allow for the plural to be interpreted inclusively in negative sentences, questions, and under the scope of a universal quantifier. Therefore, they show that languages with a plural marking system such as Akan’s,

where both the singular and plural forms are explicitly morphologically marked, can have similar semantic interpretations as languages with plural marking systems such as English's, where the plural is explicitly morphologically marked and the singular is morphologically unmarked. With the semantic interpretation of the two types of plurals clearly defined as inclusive, I suggest that there is only one plural morpheme in Akan: the plural prefix. The morpheme previously known as the plural circumfix should therefore be treated as two separate pieces of morphology: the plural prefix and a kinship suffix.

Abbreviations

DAT = dative, DET = determiner, LOC locative, MOD = modal, NC = noun class, NEG = negative, PERF = perfective, PL = plural, PST = past, REL = relativizer, SG = singular

Data availability

Anonymized data sets and scripts used for analysis can be found at https://osf.io/3n46m/?view_only=183cafd9e316442f974f8b2d741a66cf.

Ethics and consent

These experiments were originally conducted for the purpose of the author's master's thesis at the Universität Potsdam. Within the proposal, there was no need to obtain clearing of experiments by the ethics committee, as ethical approval is not required for experiments conducted with healthy adult speakers (<https://www.uni-potsdam.de/de/senat/kommissionen-des-senats/ek>). In carrying out the experiment, I adhered to standard practices regarding data protection (in compliance with UP rules on data protection) and experimental ethics.

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Competing interests

The author has no competing interests to declare.

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